

SINA ASGHARI

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SUMMARY

A highly motivated and top-performing Computer Science graduate with a demonstrated expertise in Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG). Proven ability to drive AI innovation through a research publication on Persian RAG, hands-on experience as an LLM Developer at Iran's National AI Organization, and consistent top rankings in national AI and LLM hackathons. Eager to leverage advanced skills in model optimization, fine-tuning, and end-to-end project development to contribute to challenging AI initiatives.

EDUCATION

Iran University of Science and Technology

Bachelor of Science in Computer Science

GPA: **4.0/4.0** (Final Two Years), **3.85/4.0** (Overall), Top Quartile

Ranked 4th among universities in Iran (QS).

Tehran, Iran

Oct 2021 - Sep 2025

PUBLICATIONS

Nafisi A, Asghari S, Arvenaghi MS, Shakibania H. *No Trust Issues Here: A Technical Report on the Winning Solutions for the Rayan AI Contest* December 2025. arXiv preprint. (Paper Link)

- **Compositional Image Retrieval:** Developed a system processing visual and textual inputs for retrieval with a top accuracy of 95.38%, (1st place).
- **Zero-Shot Anomaly Detection:** Designed a model detecting and localizing anomalies without prior abnormal examples, achieving 73.14% accuracy (1st place).
- **Backdoor Detection in Models:** Proposed a method for identifying backdoored neural networks, with 78% detection accuracy (2nd place).

Bourbour Hosseinbeigi S, Asghari S, Shalchian MH, Abbasi MA. *Advancing retrieval-augmented generation for Persian: Development of language models, comprehensive benchmarks, and best practices for optimization.* December 2024. (Paper Link)

- **Challenges in Low-Resource Languages:** Addresses difficulties in constructing RAG systems for Persian, focusing on its morphology and syntax.
- **Persian-Specific Models:** Introduces *MatinaRoberta* and *MatinaSRoberta*, trained on a 73.11-billion-token Persian corpus and evaluated across three datasets.
- **Evaluation and Results:** Demonstrates *MatinaSRoberta*'s superior performance, with *Llama-3.1* models excelling in domain-specific tasks.

EXPERIENCE

Quera

Track Owner – AI Competitions

Tehran, Iran

Apr 2025 – Present

- Led the design, execution, and technical direction of large-scale AI competitions for major industry partners.
- Developed end-to-end problem statements, evaluation pipelines, and baseline models aligned with partner requirements.
- Managed the full competition lifecycle, including data preparation, scoring infrastructure, participant support, and post-event analysis.

- Organized and delivered competitions in collaboration with:
 - Divar — AI Software Engineering Challenge ([link](#))
 - Torob — LLM and Agentic AI Competition ([link](#))
 - Hamrah-e Aval (MCI) — LLM Safety Challenge ([link](#))
 - Tech Olympics — Image, Text, and Data Processing Challenges ([link](#))
 - MAPNA Group — Industrial AI Hackathon ([link](#))

National Artificial Intelligence Organization of Iran, Matina Team

LLM Developer

Tehran, Iran
Apr 2024 - Jan 2025

- Collaborating on the development of an advanced LLM-based product, focusing on Persian language applications.
- Conducted model optimization, fine-tuning, and evaluation to enhance LLM performance across diverse tasks.
- Integrated retrieval-augmented generation (RAG) techniques to improve the relevance and accuracy of model outputs.
- Designed and tested pipelines for efficient data processing and model deployment in production.

Quera

Part-Time Education Mentor

Tehran, Iran
Jul 2023 - Dec 2024

- Designed hands-on projects in data science and AI to develop practical skills
- Led support teams, optimizing task assignments and team performance
- Provided mentorship in AI and data science, focusing on critical thinking and technical proficiency
- Promoted collaborative learning through peer-to-peer knowledge sharing

Quera

Part-Time Data Analyst

Tehran, Iran
Apr 2023 - Jun 2023

- Gained hands-on experience in data crawling, machine learning, databases, and Python, developing a solid foundation in data science.
- Developed interactive dashboards using Power BI, transforming complex datasets into actionable insights.
- Applied data warehousing techniques to manage and organize large datasets efficiently.
- Collaborated on two team-based projects, integrating skills in data crawling, machine learning, and data visualization to deliver comprehensive data-driven solutions.
- Enhanced proficiency in key data analysis tools and techniques, contributing to the successful completion of real-world projects.

Iran University of Science and Technology

Teaching Assistant

Tehran, Iran
Oct 2023 - Jan 2025

- Held tutorial classes, designed assignments/projects, provided support for students
- Courses: Algorithms Design, Data Structures and Algorithms, Theory of Computation, and Combinatorics

PROJECTS

Persian RAG ([GitHub Link](#))

- Project implements a Retrieval-Augmented Generation (RAG) system specifically for the Persian language, leveraging the Llamaindex library. It combines a state-of-the-art Persian language model with a highly optimized retrieval mechanism, making it one of the most effective solutions for Persian question answering tasks.
- Tools Used: Python, LlamaIndex, Langchain, HuggingFace, Commercial LLMs

Dual Task Classification: Food Images and News Articles (GitHub Link)

- Developed a classification system using deep learning models (InceptionV3, ResNet, DenseNet) for food image classification, and NLP models with AG News and BBC datasets for news article classification. Implemented techniques like data augmentation and tokenization for enhanced performance.
- Tools Used: NLP(nltk) , ML & DL (Tensorflow, Keras, Pytorch, Scikit-Learn), Visualization(wordcloud, matplotlib)

Transfermarkt Project (GitHub Link)

- Conducted data scraping from Transfermarkt to extract football transfer data, designed a database for efficient storage, performed statistical analyses, and developed machine learning models to predict player transfers and trends. Created a dashboard for data visualization and decision support.
- Tools Used: ML(Scikit-Learn), Statistics(scipy), Data Preprocessing & Visualization(Pandas, Numpy, Matplotlib, Seaborn), Dashboard (Streamlit, PowerBI)

HONORS AND AWARDS

- **University Entrance Examination:** Ranked among the top 0.1 % of students in Mathematics and Physics.
- **National ML Hackathon by MTNIRanCell:** Ranked 5th among 580 participants, Scored No.1 Contest Score
- **National LLM Hackathon By Quera:** Ranked 2nd among 1103 participants
- **RAYAN International Competition in Image Trustworthiness:** Achieved 1st place after three main phases. RAYAN Competition Link Our Technical Report
- **National AI Hackathon (Computer Vision) By Quera:** Ranked 1st among 220 participants after two phases.
- **National LLM Agent By Sharif University of Technology:** Ranked 3rd among 300 participants after three phases. LLM-Agent Competition Link
- **Top student in Computer Science Major 2025:** Ranked 3rd among 48 students.

LANGUAGES

- English – IELTS (Academic): Overall 7.0 (L: 7.5, R: 7.0, W: 6.0, S: 6.5)
- Persian – Native
- Arabic – basic
- Azari – basic

RESEARCH INTERESTS

Deep Learning, Large Language Models, Trustworthy AI, Deep Learning-Guided Program Synthesis, Reinforcement Learning, Multimodal Generation, Machine Learning

REFERENCES

- Javad Vahidi, Associate Professor & Head of the Computer Science Department, Department of Mathematics and Computer Science, Iran University of Science and Technology (IUST); jvahidi@iust.ac.ir

- Ebrahim Ardeshtir-Larijani, Assistant Professor, Department of Mathematics and Computer Science, (IUST); larijani@iust.ac.ir
- Maedeh Sadat Tahaei, Assistant Professor, Department of Mathematics and Computer Science, (IUST); mstahaei@aut.ac.ir
- Mahmoud Shirazi, Department of Computer Science and Information Technology, Institute for Advanced Studies in Basic Sciences (IASBS); m.shirazi@iasbs.ac.ir